



BEDOK SOUTH SECONDARY SCHOOL 2026 PRELIMINARY EXAMINATION

4E 5N

CANDIDATE
NAME

CLASS

INDEX
NUMBER

MATHEMATICS

Paper 2

4052/02

28 August 2026

Candidates answer on the Question Booklet.

2 hours 15 minutes

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

The number of marks is given in brackets [] at the end of each question or part question.

If working is needed for any question it must be shown with the answer.

Omission of essential working will result in loss of marks.

The total of the marks for this paper is 90.

The use of an approved scientific calculator is expected, where appropriate.

If the degree of accuracy is not specified in the question and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

For π , use either your calculator value or 3.142.

	Questions	Marks Deducted
Accuracy		-1 / -2
Presentation		
Units		

For Examiner's Use
90
Student check and sign
Parent/Guardian Signature

Mathematical Formulae*Compound Interest*

$$\text{Total amount} = P \left(1 + \frac{r}{100} \right)^n$$

Mensuration

$$\text{Curved surface area of a cone} = \pi r l$$

$$\text{Surface area of a sphere} = 4\pi r^2$$

$$\text{Volume of a cone} = \frac{1}{3} \pi r^2 h$$

$$\text{Volume of a sphere} = \frac{4}{3} \pi r^3$$

$$\text{Area of a triangle } ABC = \frac{1}{2} ab \sin C$$

$$\text{Arc length} = r\theta, \text{ where } \theta \text{ is in radians}$$

$$\text{Sector area} = \frac{1}{2} r^2 \theta, \text{ where } \theta \text{ is in radians}$$

Trigonometry

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Statistics

$$\text{Mean} = \frac{\sum fx}{\sum f}$$

$$\text{Standard deviation} = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f} \right)^2}$$

Answer **all** the questions.

- 1 (a) Expand and simplify $3(2x - 5) - 4(x + 7)$.

Answer [2]

- (b) Factorise completely $42a^2b^3 + 56ab^2$.

Answer [2]

- (c) t is directly proportional to the square of d . When $d = 2.5 \times 10^3$, $t = 9.375 \times 10^2$.

For **parts (i) and (ii)**, give your answers in standard form.

- (i) Find an equation connecting t and d .

Answer [3]

- (ii) Hence find the values of d when $t = 2 \times 10^3$.

Answer [2]

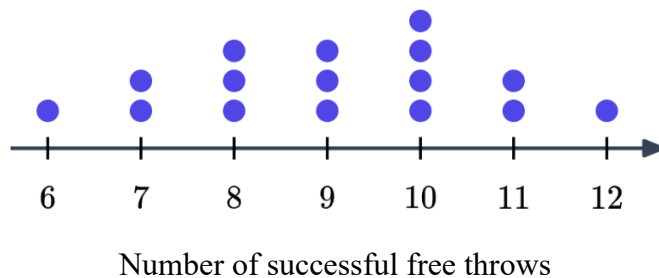
- 2 (a) Ten numbers are arranged in descending order as shown.

25, z , 18, 16, y , 11, 9, 5, x , 2

The mean and median of the ten numbers is 12. Given that the mode of the ten numbers is 18, find the values of x , y and z .

Answer $x = \dots\dots\dots$, $y = \dots\dots\dots$, $z = \dots\dots\dots$ [4]

- (b) The dot diagram shows the number of successful free throws made by 16 players. Each player attempted 15 free throws.



- (i) A player is selected at random. Find the probability that this player successfully made at least 60% of his attempts.

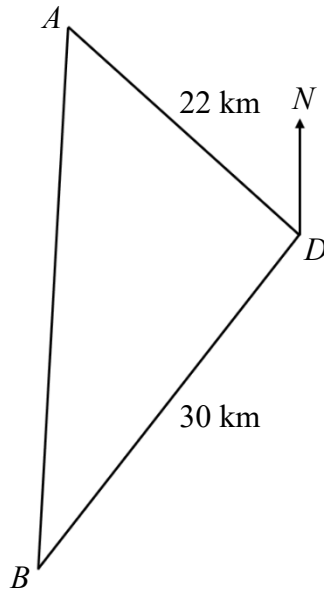
Answer $\dots\dots\dots$ [2]

A player is not selected to the team if the number of successful free throws the player made is less than m .

- (ii) Find the value of m if 13 players were selected to the team.

Answer $m = \dots\dots\dots$ [2]

- 3 A coast guard patrol ship leaves its dock at D to respond to distress signals from two boats A and B . Boats A and B are 22 km and 30 km from the dock respectively and their respective bearings from the dock are 318° and 232° .



The captain responds to boat A first, then boat B before returning to the dock. The route the patrol ship takes is outlined in triangle DAB above.

- (a) Calculate the bearing of the dock from boat B .

Answer [2]

- (b) Calculate the distance between boats A and B .

Answer km [3]

When the area of patrol exceeds 300 km^2 , the captain should seek assistance from a patrolling helicopter in the same region.

- (c) The captain thinks he should seek assistance for this operation. Do you agree?

..... [2]

- 4 The first three figures in a sequence are shown below. Each figure consists of shaded and unshaded squares. Let S_n and U_n be the number of shaded and unshaded squares in Figure n respectively.

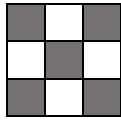


Figure 1

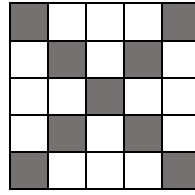


Figure 2

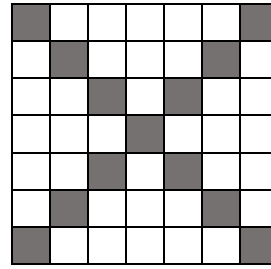


Figure 3

- (a) Write down the number of shaded squares in the next two figures.

Answer, [2]

- (b) Find a general expression, in terms of n , for the

- (i) number of shaded squares, S_n , and

Answer [2]

- (ii) number of unshaded squares, U_n .

Answer [2]

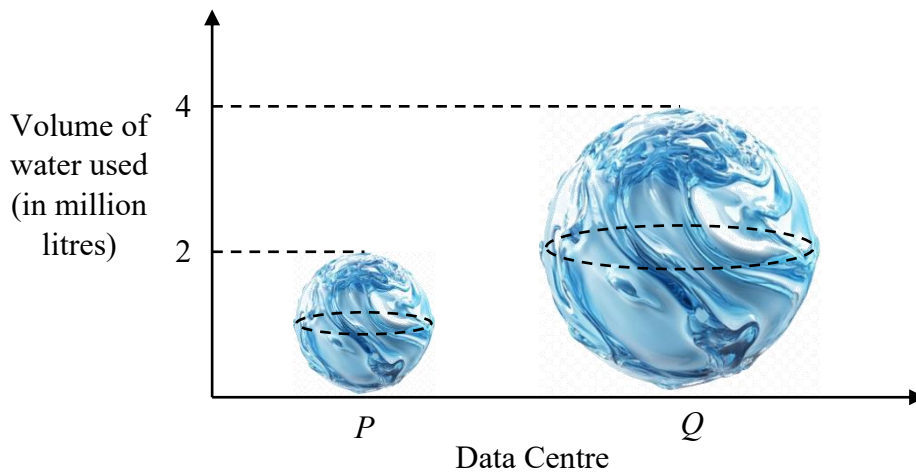
- (c) Calculate the number of unshaded squares in Figure 81.

Answer [2]

- (d) Determine, with calculations, if a figure in the sequence has 2026 shaded squares.

..... [2]

- 5 The volume of water used by two data centres to cool their systems is displayed in the chart below.



The chart shows two geometrically similar spheres representing the water usage. The dimensions of the sphere for data centre P are half the corresponding dimensions of the sphere for data centre Q .

- (a) Based on the dimensions of the spheres, calculate the ratio of the volume of water used by data centre P to that of data centre Q .

Answer [2]

- (b) Based on your answer in **part (a)**, explain why the chart may be misleading.

.....

..... [1]

A graphics designer decides to redraw the spheres in the chart to reflect the ratio of volume of water used.

- (c) If the radius of the sphere for data centre P is 3 cm, calculate the radius of the sphere for data centre Q .

Answer cm [2]

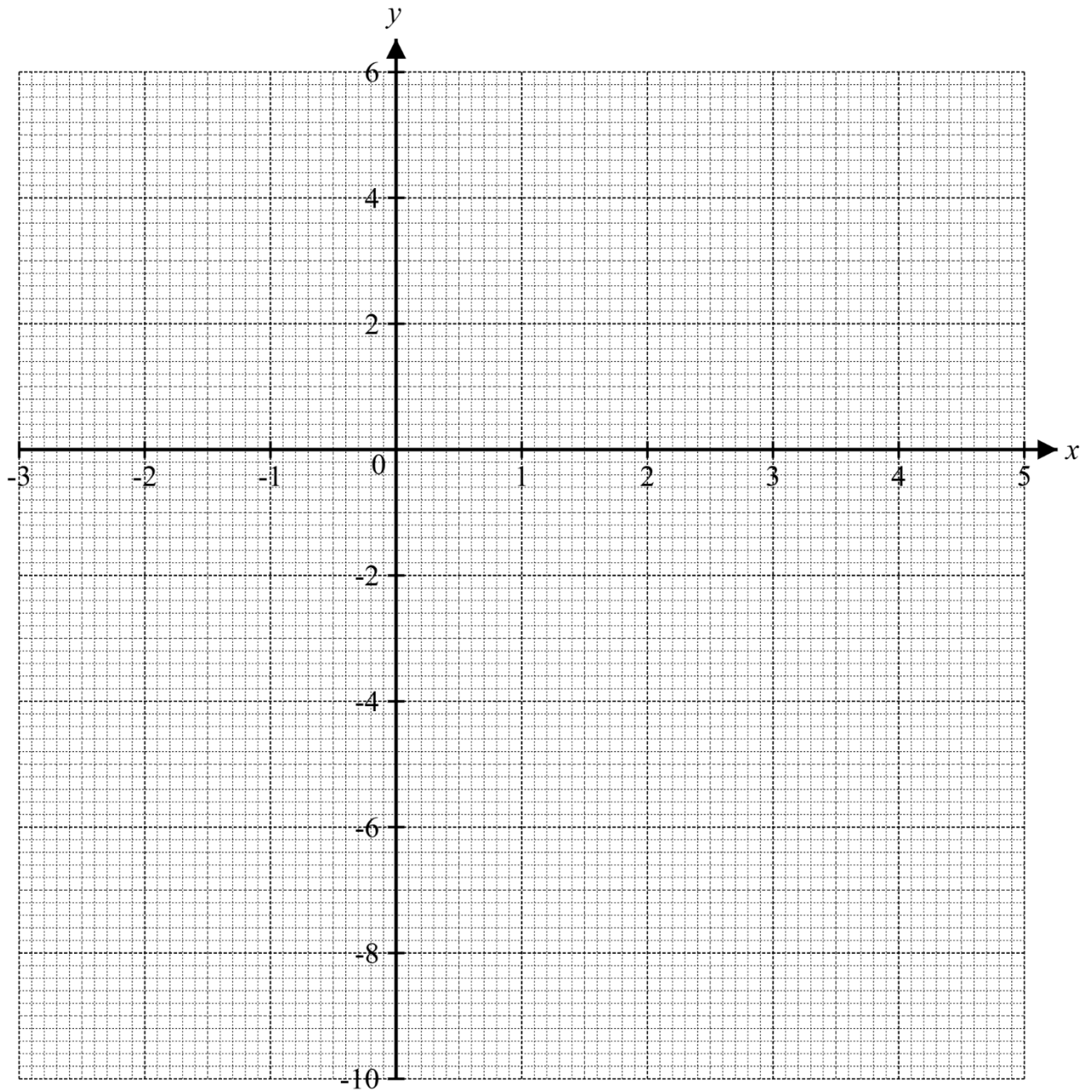
- 6 (a) Complete the table of values for $y = \frac{x^3}{2} - 2x^2 + 4$.

Values are given to one decimal place where appropriate.

x	-2	-1	0	1	2	2.5	3	3.5	4
y	-8	1.5	4	2.5	0		-0.5		4

[2]

- (b) On the grid, draw the graph of $y = \frac{x^3}{2} - 2x^2 + 4$ for $-2 \leq x \leq 4$.



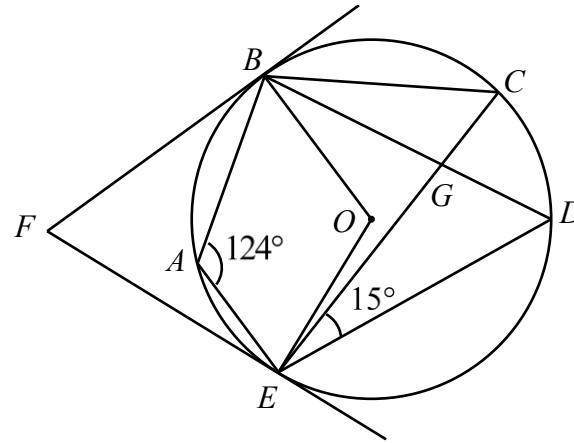
[3]

- (c) From your graph, find the range of values of k for which $\frac{x^3}{2} - 2x^2 + 4 = k$ has 3 solutions.

Answer [2]

- (d) By drawing a suitable straight line on the grid, solve the equation $x^3 - 4x^2 - 6x + 6 = 0$ for $-2 \leq x \leq 4$.

Answer $x =$ or [4]



The diagram shows a circle $ABCDE$ with, centre O . BF and EF are tangents to the circle at B and E . Angle $BAE = 124^\circ$ and angle $GED = 15^\circ$.

- (a) Find angle BOE .
Give a reason for each step of your working.

Answer [2]

- (b) Find angle BFE .
Give a reason for each step of your working.

Answer [3]

- (c) Find the sum of angle GBO and angle GEO .
Give a reason for each step of your working.

Answer [3]

- 8 An investment firm manages two portfolios, A and B, across three asset classes. The table shows the number of asset units held in each portfolio.

	Equities	Bonds	Commodities
A	500	800	200
B	300	1200	400

- (a) Represent the information in a 2×3 matrix $\mathbf{P}$.

Answer $\mathbf{P} = \begin{pmatrix} & & \\ & & \end{pmatrix}$ [1]

- (b) The matrix $\mathbf{V}$ shows the unit prices for the different assets in October.

$$\mathbf{V} = \begin{matrix} \text{Oct} \\ \begin{pmatrix} 40 \\ 15 \\ 25 \end{pmatrix} \end{matrix} \begin{matrix} \text{Equities} \\ \text{Bonds} \\ \text{Commodities} \end{matrix}$$

Evaluate the matrix $\mathbf{T} = \mathbf{PV}$.

Answer $\mathbf{T} = \begin{pmatrix} & \\ & \end{pmatrix}$ [2]

- (c) State what the elements of $\mathbf{T}$ represent.

.....
 [1]

- (d) To manage portfolio risk, financial regulators calculate a Risk Exposure Score for each class of asset using risk factors.

Equities are high risk assets with a risk factor of 1.5 per unit.

Bonds are low risk assets with a risk factor of 0.4 per unit.

Commodities are moderate risk assets with a risk factor of 1.0 per unit.

Represent these factors in a 3×1 column matrix $\mathbf{R}$.

$$\text{Answer} \quad \mathbf{R} = \begin{pmatrix} \\ \\ \end{pmatrix} \quad [1]$$

- (e) The Risk Exposure Scores of portfolios $\mathbf{E}$, is calculated by multiplying $\mathbf{R}$ and $\mathbf{P}$. A portfolio is classified as “High Risk” if its total Risk Exposure Score exceeds 1300.

By evaluating $\mathbf{E}$, determine if Portfolio A and Portfolio B are classified as “High Risk”.

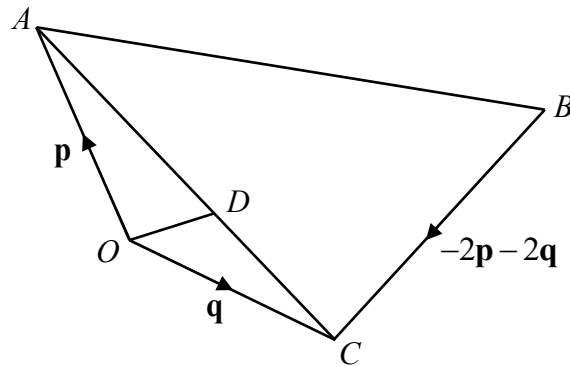
.....
 [3]

- (f) In November, the firm introduces a Reduction Factor k that is multiplied to $\mathbf{E}$. This causes the Risk Exposure Scores of both Portfolios A and B to be less than 1000.

Find the largest possible value of k , giving your answer to two decimal places.

Answer $k = \dots\dots\dots$ [2]

- 9 $OABC$ is a quadrilateral. D is a point on the diagonal AC such that $AD : DC = 3 : 2$.
 $\overrightarrow{OA} = \mathbf{p}$, $\overrightarrow{OC} = \mathbf{q}$ and $\overrightarrow{BC} = -2\mathbf{p} - 2\mathbf{q}$.



- (a) Find $\overrightarrow{OD}$. Give your answer as simply as possible in terms of $\mathbf{p}$ and $\mathbf{q}$.

Answer [4]

- (b) Explain why O , D and B lie on a straight line.

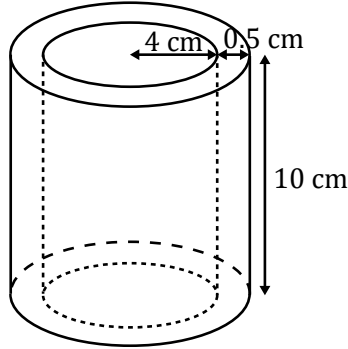
.....
 [3]

- (c) Calculate the value of $\frac{\text{Area of } \triangle OAD}{\text{Area of } OABC}$.

Answer [4]

- 10 Company M is a well-known manufacturer of cylindrical ceramic mugs based in Singapore. The company sells the mugs and provides international shipping at a cost.

Product	Dimensions	Honeycomb Wrap (if requested)
Ceramic Mug	Internal Radius = 4 cm Height = 10 cm Uniform Thickness of 0.5 cm Density of ceramic = 2.9 g/cm^3	Wrapped around the curved exterior surface



Company M receives the following order from a customer, Jerry.

<u>Order Summary</u>		
Name: <u>Jerry</u>	Order No: <u>12345</u>	
Address: <u>02929 Toronto, Canada</u>	Date: <u>8th August 2026</u>	
Product(s)	Quantity	Honeycomb Wrap
Cylindrical Mug	8	Yes
Mode of Delivery: <u>Air Postage</u>		
Mode of Payment: <u>Credit Card</u>		
Thank you for your order! A quotation for your order will be sent to you within 3 working days.		

- (a) Calculate the mass of ceramic required to produce 1 mug.

Answer g [2]

Further important information that Company M considers is provided.

Protective Material for Packaging

Items	Dimensions	Mass	Retail Price
Honeycomb Wrap	15 cm × 12 cm Uniform Thickness of 0.4 cm	5 g per piece	\$0.45 per piece

Types of Packaging Boxes

Items	Dimensions	Mass	Retail Price
Small Carton	31 cm × 23 cm × 9 cm	175 g	\$5.00
Medium Carton	40 cm × 25 cm × 15 cm	260 g	\$6.10
Large Carton	50 cm × 30 cm × 20 cm	650 g	\$7.20
Extra Large Carton	60 cm × 40 cm × 30 cm	1125 g	\$10.40

Postage Rates (International)

Destination	Air Postage		Sea Postage	
	First 2 kg	Additional kg or part thereof	First 2 kg	Additional kg or part thereof
Zone A: Malaysia	\$6.40	\$3.00	NA	NA
Zone B: Brunei, Hong Kong, Indonesia, Philippines, Taiwan and Thailand	\$12.00	\$5.00	\$7.20	\$2.00
Zone C: China, India, South Korea, rest of Asia and Oceania	\$12.00	\$5.00	\$7.20	\$2.00
Zone S: United Kingdom, Germany, Europe, USA and Canada	\$20.00	\$9.00	\$10.00	\$2.00

- (b) The mugs are to be placed upright into two rows of four in a single package. Determine the type of packaging box Company M should use to minimise cost.

Answer

- (c) Company M needs to decide on an amount to charge Jerry for this order so that it can make a profit of 25% of the total cost it incurs in fulfilling this order.

Company M incurs a cost of \$0.37 for every 100 g of ceramic used in making its mugs.

The total cost incurred includes the cost of ceramic, packaging and postage fees.

Suggest a suitable amount for Company M to charge Jerry for this order.
Justify your decision and show your calculations clearly.

.....
.....

[7]